EE605 Midsem Solutions

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1 Question 1

1.a

$\mathbb{Z}_n = \{0, 1, 2, \cdots, n-1\}$. When n is not a prime, $n = c*d$, where $1 < c, d < n$. Then $c*d \ (mod\ n) = n \ (mod\ n) = 0$. Therefore, $c * d \notin \mathbb{Z}_n \backslash \{0\}$, although $c, d \in \mathbb{Z}_n \backslash \{0\}$. Hence, $\mathbb{Z}_n$ with $mod\ n$ arithmetic is not a field when n is not a prime number.

1.b

Let the order of group G be n. p is a prime and $p|n$. Let $G_p \subseteq G$ be the set of elements of order p in G. Then the subgroup generated by $g \in G_p$ is given by $< g > = \{e, g, g^2, \cdots, g^{p-1}\}$. Since p is prime, every element of $< g > \backslash \{e\}$ has order p, and the identity element $\{e\}$ has order 1. In other words, the number of elements of order p in $< g >$ is $p - 1$.

Suppose two subgroups g and h share a non-identity element, i.e. $g^i = h^j$ for some i, j, and $g^i \neq e$. But a non-identity element of a cyclic group of prime order would generate the entire group : $< g^i >=< g >$ and $< h^j >=< h >$. Since $g^i = h^j$, therefore $< g >=< h >$. Thus no two distinct subgroups can have any non-identity element in common, and each distinct element in G_p generates a distinct subgroup of order p.

Each such subgroup (generated by an element in G_p) contributes exactly $p - 1$ distinct elements of order p. Thus the number of elements of order p in G is $(p - 1)|G_p|$, i.e. a multiple of $p - 1$.

1.c

Let G be a group of order 21. From Lagrange's theorem, the order of a subgroup of G must divide 21 (the order of group G). So G can have subgroups of order 1, 3, 7, and 21.

Consider an element p of $P = \{3, 7\}$. Since $p|21$ and p is a prime, there exists subgroup $H = \{e, h, \cdots, h^{p-1}\} = < h >$, generated by an element $h \in G$ of order $p \in P$. Thus a group of order 21 will have an element of order 3.

2 Question 2

Given binary code consists of 3 codewords : $C = \{000, 101, 111\}$. A codeword is chosen uniformly at random for transmission over a BSC. So $P(c) = \frac{1}{3} \ \forall \ c \in C$.

Error probabilities over the memoryless binary channel are as follows: $P(1|0) = 0.2, P(0|0) = 1 - 0.2 = 0.8, P(0|1) = 0.3, P(1|1) = 1 - 0.3 = 0.7$. Hence, for a transmitted word $X = x_1x_2x_3$, and a corresponding received word $Y = y_1y_2y_3$, $P(Y|X) = \prod_{i=1}^{3} P(y_i|x_i)$ (since the channel is memoryless). The received word $Y = 011$ is next decoded using the three decoding rules.

1. *Maximum Likelihood Decoding* : Let $\hat{c}$ be the decoded codeword. Then, $\hat{c} = \arg\max_{c \in C} P(Y = 011|X = c)$.

   $P(011|000) = P(0|0)P(1|0)P(1|0) = (0.8)(0.2)(0.2) = 0.032$

   $P(011|101) = P(0|1)P(1|0)P(1|1) = (0.3)(0.2)(0.7) = 0.042$

   $P(011|111) = P(0|1)P(1|1)P(1|1) = (0.3)(0.7)(0.7) = 0.147$

   Since $\max_{c \in C} P(Y|X = c) = 0.147$, $\hat{c} = 111$.

2. *Maximum Aposteriori Decoding* : $\hat{c} = \arg\max_{c \in C} P(X = c|Y = 011)$.

   Since $P(X|Y)$ is proportional to $P(X)P(Y|X)$ and $P(X = c) = \frac{1}{3}\ \forall\, c \in C$, the decoded codeword obtained using MAP decoding would be the same as that obtained using ML decoding. Therefore, $\hat{c} = 111$.

3. *Nearest Neighbour Decoding* : $\hat{c} = \arg\min_{c \in C} d_H(X = c, Y = 011)$, where d_H is the Hamming distance.

   $d_H(000, 011) = 2, d_H(101, 011) = 2, d_H(111, 011) = 1$. Therefore, $\min_{c \in C} d_H(X = c, Y = 011) = 1$, and $\hat{c} = 111$.

3 Question 3

Let us name the missing bits as $b_1, \ldots, b_6$.

$$H = \begin{pmatrix} 1 & 0 & 0 & b_1 & b_2 & 1 \\ 0 & 1 & 0 & b_3 & b_4 & 1 \\ 0 & 0 & 1 & b_5 & b_6 & 0 \end{pmatrix}$$

Now, it is given that $u = (1, 0, 0, 1, 1, 0)$ is a codeword of C. So $ur^T = 0$ for every row r of H. We get the below three equations.

$$\begin{aligned} 1 + b_1 + b_2 = 0 &\Rightarrow b_1 + b_2 = 1 \\ b_3 + b_4 &= 0 \\ b_5 + b_6 &= 0 \end{aligned}$$

Following are the possibilities: $(b_1, b_2) \in \{(0, 1), (1, 0)\}$, $(b_3, b_4) \in \{(0, 0), (1, 1)\}$, and $(b_5, b_6) \in \{(0, 0), (1, 1)\}$. So there are 8 possible values of $(b_1, \ldots, b_6)$. Writing them in matrix form will be convenient for easy elimination. That is done below.

$$B = \begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \\ b_5 & b_6 \end{pmatrix} \in \left\{ \begin{pmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \right\}$$

Now, $d(C) = 3 \Rightarrow$ some 3 columns of H must be linearly dependent, and every set of 2 columns of H must be linearly independent. So we can't have any zero-columns in H, and we also can't have repeated columns (in particular, B can't have $(1, 0, 0)^T, (0, 1, 0)^T, (0, 0, 1)^T$, or $(1, 1, 0)^T$ in its columns). This leaves us with only two alternatives:-

$$B \in \left\{ \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \right\}$$

So the only two possible values of H are

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

and in each of these cases, it is trivial to find a set of 3 columns of H which is linearly dependent.

4 Question 4

4.a

The GF(5) operations are :

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

*	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Given parity check matrix of (4, 2)-linear code C is

$$H = [-A^T|I_2] = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{bmatrix}$$

Therefore, a generator matrix of C is

$$G = [I_2|A] = \begin{bmatrix} 1 & 0 & 4 & 3 \\ 0 & 1 & 4 & 2 \end{bmatrix}$$

4.b

The minimum distance d_{min} of C can be obtained by checking the minimum weight of a non-zero codeword $c \in C$ s.t. $cH^T = 0$. Let $c = (c_1c_2c_3c_4)$ be a codeword and h_1, h_2, h_3, h_4 be the four columns of H. Then $\sum_{i=1}^{4} c_ih_i = 0$.

Note that $c_ih_i + c_jh_j \neq 0$ for any $i, j \in \{1, 2, 3, 4\}$, and $i \neq j$. This can also be verified by checking that every 2x2 submatrix of H is non-singular (i.e. det $\neq 0$), which means that any two columns of H are independent.

Taking the $1^{st}, 3^{rd}$, and 4^{th} columns, we get $c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-c_1) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-2c_1) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$ for any $c_1 \in \{1, 2, 3, 4\}$.

Similarly, taking the $2^{nd}, 3^{rd}$, and 4^{th} columns, we get $c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + (-c_2) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-3c_2) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$ for any $c_2 \in \{1, 2, 3, 4\}$.

Therefore, the minimum distance d_{min} of C is 3.

This code can detect upto $d_{min} - 1 = 2$ errors, and can correct upto $\lfloor \frac{d_{min}-1}{2} \rfloor = 1$ error.

4.c

A code is an MDS code if it satisfies the Singleton bound with equality, i.e. $d_{min} = n - k + 1$.

For given code C, $n - k + 1 = 4 - 2 + 1 = 3 = d_{min}$. Therefore, C is an MDS code.

5 Question 5

Rename T as P to avoid confusion with transpose. Let C be an $[n, k]-$ code, and let $P \subseteq \{1, \dots, n\}$. Recall that the shortening of C at P is defined as

$$C_P = \{x^P : x \in C \text{ and } x_i = 0 \, \forall \, i \in P\}$$

and the puncturing of C at P is defined as

$$C^P = \{x^P : x \in C\}$$

where x^P is the tuple of components of the codeword x with indices not in P i.e. with indices in $S = \{1, \dots, n\} \setminus P$. We will show that $(C^\perp)_P \subseteq (C^P)^\perp$ and then $(C^P)^\perp \subseteq (C^\perp)_P$, thereby showing equality. For a quick sanity check, see that atleast the lengths of these two codes match.

1. Let $v \in (C^\perp)_P \Rightarrow \exists \, v' \in C^\perp$ such that $v'_i = 0 \, \forall \, i \in P$ and $v = v'^P$. Now let $u \in C^P \Rightarrow \exists \, u' \in C$ such that $u = u'^P$. Since $v' \in C^\perp$, $u'v'^T = 0 \Rightarrow \sum_{i=1}^{n} u'_iv'_i = 0 \Rightarrow \sum_{i \in S} u'_iv'_i = 0$ since $v'_i = 0 \, \forall \, i \in P = \{1, \dots, n\} \setminus S \Rightarrow \sum_{j=1}^{n-|S|} u_jv_j = 0 \Rightarrow uv^T = 0$. Since this holds for all $u \in C^P$, $v \in (C^P)^\perp$ is forced $\Rightarrow (C^\perp)_P \subseteq (C^P)^\perp$.

2. Now say $v \in (C^P)^\perp \Rightarrow$ for all $u \in C^P$, $uv^T = 0$. Define a new vector $v' \in \mathbb{F}_q^n$ (assuming C is defined over $\mathbb{F}_q$) as

$$v'_i = \begin{cases} v_i, & \text{if } i \notin P \\ 0, & \text{otherwise} \end{cases}$$

Then for any $u' \in C$, we get $u'v'^T = \sum_{i=1}^{n} u'_iv'_i = \sum_{i \in S} u'_iv'_i = \sum_{j=1}^{n-|S|} u'^P_j v_j = 0$ since $v \in (C^P)^\perp$. Since this holds for all $u' \in C$, we must have $v' \in C^\perp$. But see that $v = v'^P$, and $v'_i = 0 \, \forall \, i \in P \Rightarrow v \in (C^\perp)_P$. So $(C^P)^\perp \subseteq (C^\perp)_P$.

6 Question 6

6.a

C is a $[6,3]_5-$MDS code $\Rightarrow d(C) = n-k+1 = 6-3+1 = 4$. Let the weight distribution of C be $\{A_i\}_{i=0}^{6}$ i.e. for $i \in \{0\ldots,6\}$, A_i is the number of codewords of C of weight i (note that weight obviously cannot be negative, and it also cannot be more than the length of the code). Then we must have $\sum\limits_{i=0}^{6} A_i = |C| = q^k = 5^3 = 125$. But $A_0 = 1$ always (since the zero codeword is an element of every linear code, and furthermore if a codeword has weight 0, it must be the zero codeword), and notice that $A_1 = A_2 = A_3 = 0$ since $d(C) = 4$ and so if a codeword of C has nonzero weight, it must have weight atleast 4. Directly, $A_4+A_5+A_6 = 125-(A_0+A_1+A_2+A_3) = 125-1 = 124$.

6.b

The dual of an $[n,k]_q-$code is an $[n,n-k]_q-$code, so $C^{\perp}$ is a $[6,3]_5-$code as well. Further, it is given that the dual of an MDS code is MDS $\Rightarrow C^{\perp}$ is MDS $\Rightarrow d(C^{\perp}) = n-(n-k)+1 = k+1 = 3+1 = 4$. Using the same logic as in C, we get that $A_1^{\perp} = A_2^{\perp} = A_3^{\perp} = 0$, where $\{A_i^{\perp}\}_{i=0}^{6}$ is the weight distribution of $C^{\perp}$. In particular, $A_1^{\perp} = A_2^{\perp} = 0$. Now we use MacWilliams identities as provided.

$$A_r^{\perp} = \frac{1}{|C|}\sum_{i=0}^{n} A_i K_r(i)$$

Put $q = 5$, $n = 6$ and $|C| = 125$.

1. For $r = 1$, we get

$$\begin{aligned} A_1^{\perp} &= \frac{1}{125}\sum_{i=0}^{6} A_i K_1(i) \\ \Rightarrow 0 &= A_0K_1(0) + A_4K_1(4) + A_5K_1(5) + A_6K_1(6) \end{aligned}$$

   since $A_1 = A_2 = A_3 = 0$. See that $K_1(i) = (q-1)(n-i) - i = (5-1)(6-i) - i = 4(6-i) - i = 24-5i$

$$\begin{aligned} &\Rightarrow 0 = 1(24-5\times 0) + A_4(24-5\times 4) + A_5(24-5\times 5) + A_6(24-5\times 6) \\ &\Rightarrow 0 = 24 + 4A_4 - A_5 - 6A_6 \end{aligned}$$

2. For $r = 2$, we similarly get

$$A_2^{\perp} = 0 = A_0K_2(0) + A_4K_2(4) + A_5K_2(5) + A_6K_2(6)$$

   As given, $K_2(i) = \binom{n-i}{2}(q-1)^2 - (n-i)i(q-1) + \binom{i}{2} = \binom{6-i}{2}4^2 - 4(6-i)i + \binom{i}{2} = 8(6-i)(5-i) - 4i(6-i) + \binom{i}{2}$. There is a subtlety here. Note that $\binom{p}{q}$ is in general equal to 0 whenever $p < q$. So $K_2(6) = \binom{6}{2} = 15$, $K_2(5) = -4\times 5 + \binom{5}{2} = -10$, $K_2(4) = 8\times 2 - 4\times 4\times 2 + \binom{4}{2} = -10$ and $K_2(0) = 8\times 6\times 5 = 240$. Therefore,

$$0 = 240 - 10A_4 - 10A_5 + 15A_6$$

6.c

Solving the three equations, we get $A_4 = 60, A_5 = 24$, and $A_6 = 40$. So we have determined the complete weight distribution of C, since we know the values of A_i for all $i \in \{0,\ldots,6\}$. But C was an arbitrary $[6,3]_5-$MDS code. Therefore, every such code must have the same weight distribution.